SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I R.Varshini/192126020T *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Varshini

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	–	–	–	–

Total Marks 32 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

SIMATS ENGINEERING

ASSESSMENT TOOL - 1

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SNO : 38

Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The transport layer divides the file into 1500-byte segments, while the Data link layer encapsulates each segments into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted overhead introduced by the data link layer, and estimate the total transmission (ignoring propagation delay). Explain how the transport and Data link layers ensure reliable communication.

File Transfer, Transmission Overhead and transmission Time.

File Size = 150MB

Bandwidth = 50 Mbps

Segment Size = 1500 bytes

Header and trailer = 40 bytes per frame

Calculate the total number of segments

Assume 1 MB = 10^6 bytes

File size = $150 \times 10^6 = 150,000,000$ bytes

Number of segments = $\frac{150,000,000}{1500}$

= 100,000 segments.

Calculate the total number of frames.

Each segments is encapsulated into one frame.

Total frames = 100,000

Calculate the transmission overhead.

Each frame has 40 bytes of header and trailer.

Overhead = $100,000 \times 40$

4,000,000 bytes = 4 MB

calculate the total transmission time

Total transmitted data:

150,000,000 + 4,000,000 = 154,000,000 bytes

Convert bytes into bits:

$154,000,000 \times 8 = 1,232,000,000$ bits

Time = $\frac{\text{Total bits}}{\text{Bandwidth}}$

$$= \frac{1,233,000,000}{50,000,000}$$

24.64 Seconds

* Transport layer : Uses Sequence numbers, acknowledgements and retransmissions to deliver data reliably. TCP also provides error detection and flow control.

* Data Link layer : Uses frame checking, such as CRC, to detect errors. Protocols may use acknowledgements and retransmissions to corrects transmissions to correct transmission errors on a link.

100,000 Segments, 100,000 frames, 4 MB overhead and 24.64 Seconds transmission time

2. A Transport layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the Sender transmits 48 frames, calculates the number of transmission window is lost and must be retransmitted, determined the total number of transmissions. Explain how flow control improves communication efficiency.

Sliding Window protocol and flow control.

* Sliding window size = 6 frames
* Data per frame = 1024 bytes
* Total frames = 48
* one frames is lost in every window.

Calculate the number of number of transmission windows.

Number of windows = $\frac{\text{Total frames}}{\text{Window size}}$

$= \frac{48}{6} = 8$ windows

Calculate the total number of retransmissions

one frame is lost in each window.

Retransmissions = 8 × 1

8 Retransmissions.

calculate the total data transmitted

Total data = 48 × 1024

= 49,152 bytes.

8 × 1024 = 8,192 bytes

49,152 + 8,192 = 57,344 bytes

8 transmission windows and 8 retransmissions

Total data sent, including retransmissions is 57,344 bytes.

2. A 12,000 byte IP packet is transmitted through a router where the maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-bytes IP header. Calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

IP fragmentation and Network Layer

* IP packet size = 12,000 bytes
* MTU = 1500 bytes.
* IP header size = 20 bytes.

Calculate the data carried by each fragment.

Data per fragment = 1500 − 20
= 1480 bytes.

Since the fragment data size must be a multiple of 8 bytes 1480 bytes is valid.

Calculate the original data size

Assuming the 12,000 byte packet includes its original 20-byte IP header.

original data = 12,000 − 20 = 11,980 bytes

Calculate the number of fragments

Number of fragments = $\left\lceil \frac{11,980}{1480} \right\rceil$

= $\lceil 8.09 \rceil$

9 fragments

The first 8 fragments carry 1480 bytes each, and the last fragment carries:

11,980 - (8 × 1480) = 140 bytes

Calculate the total header overhead

Each fragment has a 20 byte IP header

Total header overhead = 9 × 20 = 180 bytes

* Fragmentation: Divides a large IP packet into smaller fragments to fit the MTU.

* Identification: Uses an identification field to identify fragments belonging to the same original packet.

* Fragment offset: Indicates the position of each fragment in the original packet.

* Flags: Indicate whether more fragments follow.

* Reassembly: The destination combination the fragments to reconstruct the original packet.

a fragment and 180 bytes of total IP header overhead.

A multimedia application transfers a 600 MB video file. The session layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmitting, a failure occurs after 390 MB has been transferred. Determined the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the session layer.

Session layer Synchronization and checkpoints.

* Total video file size = 600 MB
* checkpoint interval = 60 MB
* Data transferred before failure = 390 MB.

calculate the number of checkpoints created

8 Number of checkpoints = $\frac{\text{Data transferred}}{\text{Checkpoint interval}}$

$= \frac{390}{60} = 6.5$

A checkpoint is created after every complete 60 MB interval.

The checkpoints occurs at 60, 120, 180, 240, 300 and 360 MB.

6 checkpoints

Calculate the data to be retransmitted.

The last checkpoint was created at 360 MB.
The failure occurred at 390 MB.

Data to retransmit = 300 - 360

30MB

After recovery, transmission resumes from the last checkpoint at 360 MB.

Importance of Synchronization in the Session Layer.

* checkpoints: Save the progress of a long data transfer.
* Recovery: Allow transmission to resume from the last checkpoint after a failure.
* Time Saving: Avoid retransmitting the entire file.
* Efficiency: Reduce unnecessary network traffic and improve resource utilization.

6 checkpoints and 30 MB of data to be retransmitted.